

NAGA SRAVANI KOMARAVOLU

📍 Herndon, VA. | [LinkedIn](#) | ✉️ nagasravani2412@gmail.com | ☎️ +1 (706)-714-0007

SUMMARY

Software Engineer skilled in building scalable, distributed systems and integrating machine learning solutions. Proficient in full-stack development, distributed computing, and delivering AI-driven applications focused on efficiency and privacy. Experienced in deep learning, data analysis, and collaborating with cross-functional teams to deliver innovative and impactful projects. Passionate about leveraging data-driven strategies to drive business growth and innovation.

SKILLS

Machine Learning: PyTorch, NumPy, Pandas, Differential Privacy (DP), Scikit-learn, LLMs, DGL, RDKit, Matplotlib, Seaborn, GNNs, CNNs, RNNs, K-NN, Decision Trees, CUDA, Spark, Tableau, Excel

Programming: Python, Java, C++, MySQL, HTML/CSS, MongoDB, PHP, JavaScript, C, Django, React, GIT, Airflow, Docker, Jenkins, Kubernetes, RESTful, PostgreSQL.

Certification: Certified AWS Solution Architect, Google Professional Machine Learning Engineer, Cisco CCNA

EXPERIENCE

Software Engineer, 'MyYouthSpan' AI Integration and Backend Optimization 06/2024 - Present

MyEdMaster R&D – Reston, Virginia, United States

- Currently integrating an AI assistant into the platform and leveraging AWS SageMaker to deliver personalized health reports, improving user engagement through tailored recommendations.
- Designed and implemented RESTful APIs for real-time data collection and secure model training, reducing latency by 19% and ensuring zero raw-data transfer across devices, focusing on data scalability.
- Developed modular backend components using Django and Python to support the AI assistant for the social media platform of 'MyYouthSpan' website, enhancing system efficiency and achieving a 15% accuracy boost.
- Built and optimized data storage solutions using PostgreSQL reducing query times by 7%.

Research Assistant, Privacy Preserving Machine Learning Lab,

04/2023 - 05/2024

University of Georgia – Athens, Georgia, United States

- Designed and implemented an end-to-end ETL pipeline to manage and preprocess large image datasets, automating data extraction, applying data augmentation and normalization techniques, and ensuring efficient data loading.
- Trained neural network architectures, including ResNet, ConvMixer, and DenseNet, using PyTorch, and analyzed baseline performance metrics to assess the impact of differential privacy on training dynamics and model performance.
- Designed and developed an advanced ConvNet architecture with varied depth and width, resulting in a 23% improvement in model accuracy across diverse datasets through rigorous analysis of learning patterns.
- Implemented differential privacy training protocols using Opacus and TAN methods, enhancing data security by reducing data exposure while maintaining strong model performance and optimizing training efficiency.

Java Developer, TCS BANCS,

05/2021 - 08/2022

Tata Consultancy Services (TCS) – Bangalore, Karnataka, India

- Developed and maintained backend systems for banking applications using Java and JavaScript, processing over 100,000 daily transactions with a focus on reliability and scalability.
- Designed and optimized SQL queries and relational database models to streamline transaction workflows, improving data processing efficiency by 20%.
- Implemented real-time metadata-driven notification features using JavaScript and SQL, enhancing system performance and reducing response times for over 500 accounts by 15%.
- Upgraded legacy systems by adopting modern JavaScript frameworks like React for front-end modernization, improving user experience and aligning with backend microservice architecture.
- Collaborated in Agile sprints with cross-functional teams to deliver secure, high-performance banking solutions.

Machine Learning Intern,

04/2019 - 10/2019

Codemia – Hyderabad, Telangana, India

- Specialized in Machine Learning with focus on real-time applications and algorithm implementation.
- Streamlined an accurate heart disease prediction model using advanced algorithms and research, achieving an 80% accuracy rate in predicting heart disease risk in patients, leading to early intervention and improved patient outcomes.
- Utilized professional statistical techniques to maintain large databases, collecting and analyzing data from 5000 customers, resulting in actionable insights and a 20% increase in customer satisfaction scores.

EDUCATION

Master of Science, Computer Science, University of Georgia

• Athens, Georgia, United States • May 2024

PROJECTS

Automotive Assistant Voice Chatbot

- Designed a voice-activated chatbot for car showrooms to engage customers, gather preferences, provide product comparisons, and schedule sales appointments.
- Integrated advanced NLP using APIs (Google, Microsoft, OpenAI) or custom models to handle inquiries for new and used cars, enhancing customer experience and streamlining interactions.